Title: Problem Sheet 1

Course: AERO40006 (Ordinary Differential Equations)

Lecturer: Peter Vincent

Question 1: Identify the order of the following ODEs, also identify whether they are linear or non-linear.

(a) $\cos(2x)\dfrac{\mathrm{d}^4y}{\mathrm{d}x^4} + x = y$

(b) $\left(\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}\right)\left(\dfrac{\mathrm{d}y}{\mathrm{d}x}\right) = 1$

(c) $(x^2-1)\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} - \sin(x)\dfrac{\mathrm{d}y}{\mathrm{d}x} + xy = x^2$

(d) $\dfrac{\mathrm{d}y}{\mathrm{d}x} + y = x + y^2$

(a) 4th order
linear

(b) 2nd order
non-linear

(c) 2nd order
linear

(d) 1st order
non-linear

Question 2: Find particular solutions of the following simple first-order ODEs

(a) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = x \qquad y(0) = 2$

(b) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = e^x \qquad y(0) = 3$

(c) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \cos(x) \qquad y(0) = 1$

(d) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \tan(x) \qquad y(0) = 4$

(a) $\int dy = \int x\,dx$

$y = \frac{1}{2}x^2 + C$

When $y(0) = 2$,

$2 = 0 + C$

$\therefore C = 2$

$\therefore y = \frac{1}{2}x^2 + 2$

(b) $\int dy = \int e^x\,dx$

$y = e^x + C$

When $y(0) = 3$

$1 = 1 + C$

$\therefore C = 2$

$\therefore y = e^x + 2$

Question 2: Find particular solutions of the following simple first-order ODEs

(a) $\frac{\mathrm{d}y}{\mathrm{d}x} = x \qquad y(0) = 2$

(b) $\frac{\mathrm{d}y}{\mathrm{d}x} = e^x \qquad y(0) = 3$

(c) $\frac{\mathrm{d}y}{\mathrm{d}x} = \cos(x) \qquad y(0) = 1$

(d) $\frac{\mathrm{d}y}{\mathrm{d}x} = \tan(x) \qquad y(0) = 4$

(c) $\int dy = \int \cos(x)\,dx$

$y = \sin(x) + C$

When $y(0) = 1$,

$1 = 0 + C$

$\therefore C = 1$

$\therefore y = \sin(x) + 1$

(d) $\int dy = \int \tan(x)\,dx$

$= \int \frac{\sin(x)}{\cos(x)}\,dx$

Let $u = \cos(x)$,

$\frac{du}{dx} = -\sin(x)$

$\therefore dx = \frac{du}{-\sin(x)}$

$\therefore y = \int -\frac{1}{u}\,du$

$= -\ln|u| + C$

$= -\ln|\cos(x)| + C$

When $y(0) = 4$,

$4 = 0 + C \qquad \therefore C = 4$

$\therefore y = -\ln|\cos(x)| + 4$

Question 3: Find particular solutions of the following separable first-order ODEs

(a) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = (1+x)(1+y) \qquad y(0) = 0$

(b) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{1+y}{2+x} \qquad y(0) = 1$

(c) $x\dfrac{\mathrm{d}y}{\mathrm{d}x} = y + xy \qquad y(1) = 1$

(d) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = 8x^3y^2 \qquad y(2) = 3$

(a) $\int \frac{1}{1+y}\,dy = \int (1+x)\,dx$

$\ln|1+y| = x + \frac{1}{2}x^2 + A$

When $y(0) = 0$,

$A = 0$

$\therefore \ln|1+y| = x + \frac{1}{2}x^2$

$\therefore 1+y = e^{x+\frac{1}{2}x^2}$

$= e^x e^{\frac{1}{2}x^2}$

$\therefore y = e^x e^{\frac{1}{2}x^2} - 1$

(b) $\int \frac{1}{1+y}\,dy = \int \frac{1}{2+x}\,dx$

$\ln|1+y| = \ln|2+x| + A$

When $y(0) = 1$,

$A = 0$

$\therefore \ln|1+y| = \ln|2+x|$

$\therefore e^{\ln|1+y|} = e^{\ln|2+x|}$

$\therefore |1+y| = |2+x|$

$\therefore |y| = |x+1|$

$\therefore y = x+1$

Question 3: Find particular solutions of the following separable first-order ODEs

(a) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = (1+x)(1+y) \qquad y(0) = 0$

(b) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{1+y}{2+x} \qquad y(0) = 1$

(c) $x\dfrac{\mathrm{d}y}{\mathrm{d}x} = y + xy \qquad y(1) = 1$

(d) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = 8x^3y^2 \qquad y(2) = 3$

(c) $\int \frac{1+x}{x}\,dx = \int \frac{1}{y}\,dy$

$\ln|y| = \ln|x| + x + C$

When $y(1) = 1$,

$C = -1$

$\therefore \ln|y| = \ln|x| + x - 1$

$\therefore e^{\ln|y|} = e^{\ln|x| + x - 1}$

$\therefore |y| = e^{\ln|x|}e^{x}e^{-1}$

$= |x|e^{x}e^{-1}$

$\therefore y = xe^{(x-1)}$

(d) $\int \frac{1}{y^2}\,dy = \int 8x^3\,dx$

$-\frac{1}{y} = 2x^4 + C$

When $y(2) = 3$

$-\frac{1}{3} = 32 + C$

$\therefore C = -\frac{97}{3}$

$\therefore y = \dfrac{-1}{2x^4 - \frac{97}{3}}$

$\therefore y = \dfrac{-3}{6x^4 - 97}$

Question 4: Find particular solutions of the following homogeneous first-order ODEs

(a) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{x^2+y^2}{xy}$ $y(1)=0$

(b) $2x^2\dfrac{\mathrm{d}y}{\mathrm{d}x} = x^2+y^2$ $y(1)=0$

(c) $x\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{y^2}{x}+y$ $y(1)=1$

(d) $x^3\dfrac{\mathrm{d}y}{\mathrm{d}x} = x^2y-2y^3$ $y(1)=6$

(a) Let $u = \frac{y}{x}$ $\therefore y = ux$ $\therefore \frac{dy}{dx} = \frac{du}{dx}x + u$

$\therefore \frac{dy}{dx} = \frac{x}{y} + \frac{y}{x} = \frac{1}{u} + u = x\frac{du}{dx} + u$

$\therefore x\frac{du}{dx} = \frac{1}{u}$

$\therefore \int \frac{1}{x}dx = \int u\,du$

$\therefore \ln|x| = \frac{1}{2}u^2 + C$

$\therefore \ln|x| = \frac{1}{2}(\frac{y}{x})^2 + C$

When $y(1) = 0$, $C = 0$

$\therefore \ln|x| = \frac{1}{2}(\frac{y}{x})^2 = \frac{y^2}{2x^2}$

$\therefore y^2 = 2x^2\ln|x|$

$\therefore y = x\sqrt{2\ln|x|} = x\sqrt{\ln(x^2)}$

Question 4: Find particular solutions of the following homogeneous first-order ODEs

(a) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{x^2+y^2}{xy} \qquad y(1) = 0$

(b) $2x^2\dfrac{\mathrm{d}y}{\mathrm{d}x} = x^2 + y^2 \qquad y(1) = 0$

(c) $x\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{y^2}{x} + y \qquad y(1) = 1$

(d) $x^3\dfrac{\mathrm{d}y}{\mathrm{d}x} = x^2y - 2y^3 \qquad y(1) = 6$

(b) $\dfrac{dy}{dx} = \dfrac{1}{2} + \dfrac{1}{2}\left(\dfrac{y}{x}\right)^2$

Let $u = \dfrac{y}{x}$ $\quad \therefore y = ux \quad \therefore \dfrac{dy}{dx} = x\dfrac{du}{dx} + u$

$\therefore x\dfrac{du}{dx} + u = \dfrac{1}{2} + \dfrac{1}{2}u^2$

$\therefore \displaystyle\int \frac{2}{u^2 - 2u + 1}\,du = \int \frac{1}{x}\,dx$

$\therefore \displaystyle\int \frac{2}{(u-1)^2}\,du = \ln|x| + A$

$\therefore \dfrac{-2}{(u-1)} = \ln|x| + B$

$\therefore \dfrac{-2}{(\frac{y}{x}-1)} = \ln|x| + B$

When $y(1) = 0$, $B = 2$

$\therefore -2 = \dfrac{y}{x}\ln|x| - \ln|x| + \dfrac{2y}{x} - 2 \qquad \therefore x\ln|x| = y(\ln|x| + 2)$

$\cancel{\therefore -2 = \ln|x| - \frac{2y}{x} - 2 \qquad \therefore -2x = x\ln|x| - 2y - 2}$

$\cancel{\therefore y = \frac{1}{2}x\ln|x| + x - 1} \qquad \therefore y = \dfrac{x\ln|x|}{\ln|x| + 2}$

Question 4: Find particular solutions of the following homogeneous first-order ODEs

(a) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{x^2+y^2}{xy}$ $\quad y(1) = 0$

(b) $2x^2\dfrac{\mathrm{d}y}{\mathrm{d}x} = x^2 + y^2$ $\quad y(1) = 0$

(c) $x\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{y^2}{x} + y$ $\quad y(1) = 1$

(d) $x^3\dfrac{\mathrm{d}y}{\mathrm{d}x} = x^2y - 2y^3$ $\quad y(1) = 6$

(c) $\dfrac{dy}{dx} = \dfrac{y^2}{x^2} + \dfrac{y}{x}$

Let $u = \dfrac{y}{x}$ $\quad \therefore y = ux \quad \therefore \dfrac{dy}{dx} = \dfrac{du}{dx}x + u$

$\therefore x\dfrac{dy}{dx} + u = u^2 + u$

$\therefore x\dfrac{dy}{dx} = u^2$

$\therefore \int \dfrac{1}{u^2}du = \int \dfrac{1}{x}dx$

$\therefore -\dfrac{1}{u} = \ln|x| + C$

$\therefore -\dfrac{x}{y} = \ln|x| + C$

When $y(1) = 1$, $C = -1$ $\quad \therefore x = (1 - \ln|x|)y$

$\therefore y = \dfrac{x}{1 - \ln|x|}$

Question 4: Find particular solutions of the following homogeneous first-order ODEs

(a) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{x^2+y^2}{xy} \qquad y(1) = 0$

(b) $2x^2\dfrac{\mathrm{d}y}{\mathrm{d}x} = x^2+y^2 \qquad y(1) = 0$

(c) $x\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{y^2}{x} + y \qquad y(1) = 1$

(d) $x^3\dfrac{\mathrm{d}y}{\mathrm{d}x} = x^2y - 2y^3 \qquad y(1) = 6$

(d) $\dfrac{dy}{dx} = \dfrac{y}{x} - 2(\dfrac{y}{x})^3$

Let $u = \dfrac{y}{x} \quad \therefore y = ux \quad \therefore \dfrac{dy}{dx} = \dfrac{du}{dx}x + u$

$\therefore \dfrac{du}{dx}x + u = u - 2u^3$

$\therefore \dfrac{du}{dx}x = -2u^3$

$\therefore \int -\dfrac{1}{2u^3}\,du = \int \dfrac{1}{x}\,dx$

$\therefore \dfrac{1}{4u^2} = \ln|x| + C$

$\therefore \dfrac{x^2}{4y^2} = \ln|x| + C$

When $y(1) = 6$, $C = \dfrac{1}{144} \quad \therefore \dfrac{x^2}{4} = (\ln|x| + \dfrac{1}{144})y^2$

$\therefore x^2 = (4\ln|x| + \dfrac{1}{36})y^2 \quad \therefore y = \dfrac{x}{2\sqrt{\ln|x| + \frac{1}{144}}}$

Question 5: Find particular solutions of the following linear first-order ODEs

(a) $\dfrac{\mathrm{d}y}{\mathrm{d}x} + 5y = e^{2x} \qquad y(0) = 0$

(b) $x\dfrac{\mathrm{d}y}{\mathrm{d}x} - 4y - x^7 = 0 \qquad y(1) = 0$

(c) $\dfrac{\mathrm{d}y}{\mathrm{d}x} + \dfrac{y}{x} = 3x^2 \qquad y(1) = 5 \qquad x > 0$

(a) Let $I = e^{\int 5\,dx} = e^{5x}$ $\quad \therefore \dfrac{dI}{dx} = 5e^{5x}$

$$e^{5x}\frac{dy}{dx} + 5e^{5x}y = e^{5x}e^{2x} = e^{7x}$$

$$\therefore I\frac{dy}{dx} + \frac{dI}{dx}y = e^{7x}$$

$$\therefore \frac{dIy}{dx} = e^{7x}$$

$$\therefore Iy = \int e^{7x}\,dx = \frac{1}{7}e^{7x} + C \qquad \therefore e^{5x}y = \frac{1}{7}e^{7x} + C$$

When $y(0) = 0$

$$\therefore C = -\frac{1}{7}$$

$$\therefore e^{5x}y = \frac{1}{7}e^{7x} - \frac{1}{7}$$

$$\therefore y = \frac{1}{7}e^{2x} - \frac{1}{7}e^{-5x}$$

Question 5: Find particular solutions of the following linear first-order ODEs

(a) $\dfrac{\mathrm{d}y}{\mathrm{d}x} + 5y = e^{2x}$ $\quad y(0) = 0$

(b) $x\dfrac{\mathrm{d}y}{\mathrm{d}x} - 4y - x^7 = 0$ $\quad y(1) = 0$

(c) $\dfrac{\mathrm{d}y}{\mathrm{d}x} + \dfrac{y}{x} = 3x^2$ $\quad y(1) = 5$ $\quad x > 0$

(b) $\dfrac{dy}{dx} - 4\dfrac{y}{x} - x^6 = 0$

$I = e^{\int -\frac{4}{x}dx} = e^{-4\ln|x|} = e^{\ln|x|^{-4}} = x^{-4}$

$\therefore \dfrac{dI}{dx} = -4x^{-5}$

$\therefore x^{-4}\dfrac{dy}{dx} - 4x^{-5}y = x^2$

$\therefore I\dfrac{dy}{dx} - \dfrac{dI}{dx}y = x^2$

$\therefore \dfrac{dIy}{dx} = x^2$

$\therefore Iy = \int x^2 dx \quad \therefore Iy = \frac{1}{3}x^3 + C \quad \therefore x^{-4}y = \frac{1}{3}x^3 + C$

When $y(1) = 0$, $C = -\frac{1}{3}$

$\therefore x^{-4}y = \frac{1}{3}x^3 - \frac{1}{3}$

$\therefore y = \frac{1}{3}x^7 - \frac{1}{3}x^4$

Question 5: Find particular solutions of the following linear first-order ODEs

(a) $\dfrac{\mathrm{d}y}{\mathrm{d}x} + 5y = e^{2x} \qquad y(0) = 0$

(b) $x\dfrac{\mathrm{d}y}{\mathrm{d}x} - 4y - x^7 = 0 \qquad y(1) = 0$

(c) $\dfrac{\mathrm{d}y}{\mathrm{d}x} + \dfrac{y}{x} = 3x^2 \qquad y(1) = 5 \qquad x > 0$

(c) $I = e^{\int \frac{1}{x} dx} = e^{\ln|x|} = x$

$\therefore \dfrac{dI}{dx} = 1$

$\therefore x\dfrac{dy}{dx} + y = 3x^3$

$\therefore I\dfrac{dy}{dx} + \dfrac{dI}{dx}y = 3x^3$

$\therefore \dfrac{dIy}{dx} = 3x^3$

$\therefore Iy = \int 3x^3 dx = \frac{3}{4}x^4 + C$

$\therefore y = \frac{3}{4}x^3 + \frac{C}{x}$

When $y(1) = 5$, $5 = \frac{3}{4} + C \quad \therefore C = \frac{17}{4}$

$\therefore y = \frac{3}{4}x^3 + \frac{17}{4x}$

Question 6: Find general implicit solutions of the following exact first-order ODEs

(a) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{-\cos(xy) + xy\sin(xy)}{-x^2\sin(xy) + 2y}$ (b) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{3x^2 - 2xy}{x^2 - 2y}$

(a) $f(x, y) = -\cos(xy) + xy\sin(xy)$

$g(x, y) = x^2\sin(xy) - 2y$

$\because \dfrac{df}{dy} = x\sin(xy) + x\sin(xy) + x^2y\cos(xy) = 2x\sin(xy) + x^2y\cos(xy)$

$\dfrac{dg}{dx} = 2x\sin(xy) + x^2y\cos(xy)$ $\therefore \dfrac{df}{dy} = \dfrac{dg}{dx}$

Let $\dfrac{du}{dx} = f$, $\dfrac{du}{dy} = g$

$\therefore u_1 = \int f\,dx = \dfrac{1}{y}\sin(xy) - x\cos(xy) - \dfrac{1}{y}\sin(xy) + p(y)$

$= -x\cos(xy) + p(y)$

$u_2 = \int g\,dy = -x\cos(xy) - y^2 + q(y)$

$\therefore \dfrac{du_1}{dy} = x^2\sin(xy) + \dfrac{dp(y)}{dy} = x^2\sin(xy) - 2y$

$\therefore p(y) = \int -2y\,dy = -y^2 + A$

Question 6: Find general implicit solutions of the following exact first-order ODEs

(a) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{-\cos(xy) + xy\sin(xy)}{-x^2\sin(xy) + 2y}$ (b) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{3x^2 - 2xy}{x^2 - 2y}$

$\therefore \dfrac{du_2}{dx} = -\cos(xy) + xy\sin(xy) + \dfrac{dq(y)}{dx} = xy\sin(xy) - \cos(xy)$

$\therefore q(y) = 0$

$\therefore u = -x\cos(xy) - y^2 + C$

Question 6: Find general implicit solutions of the following exact first-order ODEs

(a) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{-\cos(xy) + xy\sin(xy)}{-x^2\sin(xy) + 2y}$ (b) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{3x^2 - 2xy}{x^2 - 2y}$

(b) $f = 3x^2 - 2xy$

$g = -x^2 + 2y$

$\therefore \frac{df}{dy} = -2x$, $\frac{dg}{dx} = -2x$ $\therefore \frac{df}{dy} = \frac{dg}{dx}$

$\therefore f = \frac{du_1}{dx}$, $g = \frac{du_2}{dy}$

$\therefore u_1 = \int 3x^2 - 2xy \, dx = x^3 - x^2y + p$

$u_2 = \int -x^2 + 2y \, dy = -x^2y + y^2 + q$

$\therefore \frac{du_1}{dy} = -x^2 + \frac{dp}{dy} = -x^2 + 2y$ $\therefore p = \int 2y \, dy = y^2 + A$

$\therefore \frac{du_2}{dx} = -2xy + \frac{dq}{dx} = 3x^2 - 2xy$ $\therefore q = \int 3x^2 dx = x^3 + B$

$\therefore u = x^3 + y^2 - x^2y + A + B = x^3 + y^2 - x^2y + C$

Question 7: Find general implicit solutions of the following inexact first-order ODEs. Clue: in both cases the appropriate integrating factor is a function of either x or y alone.

(a) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{x - y^2}{2y}$ (b) $(3xy - 6x^2)\dfrac{\mathrm{d}y}{\mathrm{d}x} = (6xy - y^2)$

(a) $f = x - y^2$, $g = -2y$ $\quad \therefore \frac{df}{dy} = -2y$, $\frac{dg}{dx} = 1$ $\quad \therefore \frac{df}{dy} \neq \frac{dg}{dx}$

Let $I = e^x$

$fI = xe^x - y^2e^x$, $gI = -2ye^x$ $\quad \therefore \frac{dfI}{dy} = -2ye^x$, $\frac{dgI}{dx} = -2ye^x$

$\therefore \frac{dfI}{dy} = \frac{dgI}{dx}$

Let $\frac{du_1}{dx} = fI$, $\frac{du_2}{dy} = gI$

$u_1 = \int fI\,dx = \int xe^x - y^2e^x\,dx = xe^x - e^x - y^2e^x + p$

$u_2 = \int gI\,dy = \int -2ye^x\,dy = -y^2e^x + q$

$\therefore \frac{du_1}{dy} = -2ye^x + \frac{dp}{dy} = -2ye^x \quad \therefore p = A$

$\therefore \frac{du_2}{dx} = -y^2e^x + \frac{dq}{dx} = xe^x - y^2e^x$

$\therefore q = \int xe^x\,dx = xe^x - e^x + B$

$\therefore u = xe^x - e^x - y^2e^x + C$

Question 7: Find general implicit solutions of the following inexact first-order ODEs. Clue: in both cases the appropriate integrating factor is a function of either x or y alone.

(a) $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x - y^2}{2y}$ (b) $(3xy - 6x^2)\frac{\mathrm{d}y}{\mathrm{d}x} = (6xy - y^2)$

(b) $f = 6xy - y^2 \quad \therefore \frac{df}{dy} = 6x - 2y$

$g = 6x^2 - 3xy \quad \therefore \frac{dg}{dx} = 12x - 3y \qquad \because \frac{df}{dy} \neq \frac{dg}{dx}$

Let $I = y$

$\therefore fI = 6xy^2 - y^3 \quad \therefore \frac{dfI}{dy} = 12xy - 3y^2$

$\therefore gI = 6x^2y - 3xy^2 \quad \therefore \frac{dgI}{dx} = 12xy - 3y^2 \qquad \therefore \frac{dfI}{dy} = \frac{dgI}{dx}$

$\therefore u_1 = \int fI\, dx = 3x^2y^2 - xy^3 + p$

$u_2 = \int gI\, dy = 3x^2y^2 - xy^3 + q$

$\therefore \frac{du_1}{dy} = 6x^2y - 3xy^2 + \frac{dp}{dx} = 6x^2y - 3xy^2 \qquad \therefore p = A$

$\therefore \frac{du_2}{dx} = 6xy^2 - y^3 + \frac{dq}{dx} = 6xy^2 - y^3 \qquad \therefore q = B$

$\therefore u = 3x^2y^2 - xy^3 + C$

Question 8: The population of a given species of fish in a pond is governed by the following ODE

$$\frac{\mathrm{d}N}{\mathrm{d}t} = \alpha(m - N)N,$$

where $N = N(t)$ is the number of fish at time t, α is a positive constant, and m is a positive constant known as the 'carrying capacity'. Populations below the carrying capacity have an excess of resources and hence grow, whereas populations above the carrying capacity have a shortage of resources, and hence shrink. Find a general solution for the population of fish as a function of time. Now, assuming $\alpha = 1$ and $m = 1000$, sketch plots of $N(t)$ against t for cases where $N(0) = 10$ and $N(0) = 1200$. What will always happen as $t \to \infty$. Does this make sense?

$$\int \frac{1}{(m-N)N}\,dN = \int \alpha\,dt$$

For $\int \frac{1}{(m-N)N}\,dN$. $\int \frac{1}{(m-N)N}\,dN = \int \frac{A}{m-N} - \frac{B}{N}\,dN$

$\therefore AN - Bm + BN = 1 \quad \therefore B = -\frac{1}{m} \quad \therefore A = \frac{1}{m}$

$$\therefore \int \frac{1}{m^2-mN}\,dN + \int \frac{1}{mN}\,dN = \alpha t + A$$

$$\therefore \int \frac{1}{m-N}\,dN + \int \frac{1}{N}\,dN = m\alpha t + A$$

$$\therefore -\ln|m-N| + \ln|N| + B = \ln\left|\frac{N}{m-N}\right| + B = m\alpha t + A$$

$$\therefore \left|\frac{N}{m-N}\right| = e^{m\alpha t} + C \qquad \therefore \frac{N}{m-N} = e^{m\alpha t} + C$$

Question 8: The population of a given species of fish in a pond is governed by the following ODE

$$\frac{\mathrm{d}N}{\mathrm{d}t} = \alpha(m - N)N,$$

where $N = N(t)$ is the number of fish at time t, α is a positive constant, and m is a positive constant known as the 'carrying capacity'. Populations below the carrying capacity have an excess of resources and hence grow, whereas populations above the carrying capacity have a shortage of resources, and hence shrink. Find a general solution for the population of fish as a function of time. Now, assuming $\alpha = 1$ and $m = 1000$, sketch plots of $N(t)$ against t for cases where $N(0) = 10$ and $N(0) = 1200$. What will always happen as $t \to \infty$. Does this make sense?

$\therefore \left|\frac{N}{m} - 1\right| = e^{m\alpha t} + C \qquad \therefore |N - m| = me^{m\alpha t} + C$

When $\alpha = 1$, $m = 1000$ $\qquad \therefore |N - 1000| = 1000e^{1000t} + C$

When $N(0) = 10$ $\qquad \therefore C = -10$

When $N(0) = 1200$ $\qquad \therefore C = -800$

As $t \to \infty$, $|N - 1000| \to \infty \qquad \therefore N \to \infty$

Which not make sense

$\therefore N = me^{m\alpha t} + mC - Ne^{m\alpha t} - NC \qquad \therefore N(1 + e^{m\alpha t} + C) = me^{m\alpha t} + mC$

$$\therefore N = \frac{me^{m\alpha t} + mC}{1 + e^{m\alpha t} + C}$$

Question 8: The population of a given species of fish in a pond is governed by the following ODE

$$\frac{\mathrm{d}N}{\mathrm{d}t} = \alpha(m - N)N,$$

where $N = N(t)$ is the number of fish at time t, α is a positive constant, and m is a positive constant known as the 'carrying capacity'. Populations below the carrying capacity have an excess of resources and hence grow, whereas populations above the carrying capacity have a shortage of resources, and hence shrink. Find a general solution for the population of fish as a function of time. Now, assuming $\alpha = 1$ and $m = 1000$, sketch plots of $N(t)$ against t for cases where $N(0) = 10$ and $N(0) = 1200$. What will always happen as $t \to \infty$. Does this make sense?

When $\alpha = 1$, $m = 1000$ $\quad \therefore N = \dfrac{1000e^{1000t} + 1000C}{1 + e^{1000t} + C}$

When $N(0) = 10$ $\quad \therefore C = -\dfrac{98}{99}$

When $N(0) = 1200$ $\quad \therefore C = -700$

As $t \to \infty$, $N \to 1000$ $\quad \therefore N \to m$

Which make sense